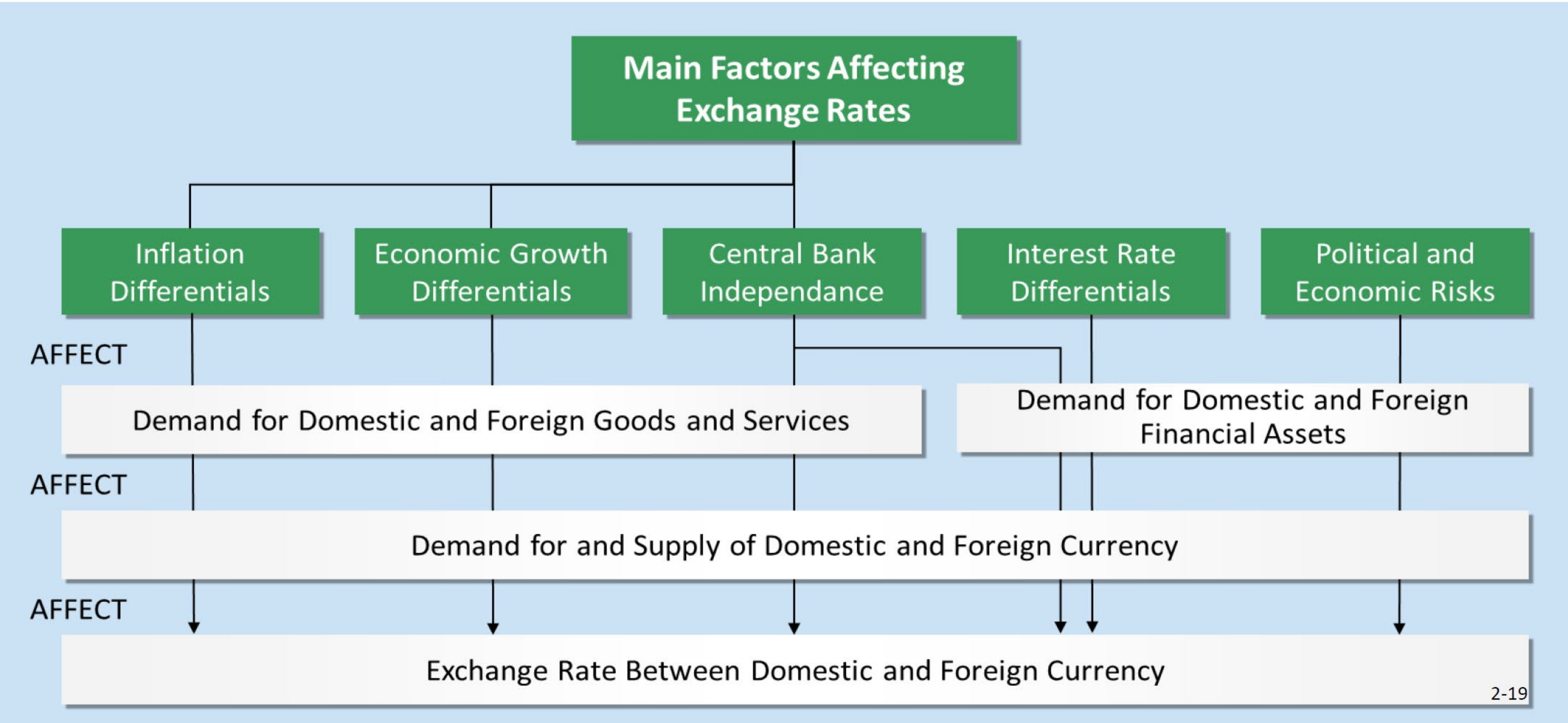
# Chapter 4: Extra proofs and comments—Part 2

**NOTE:**

1. These notes do not contain any new examinable material.
2. They contain some elaborations on the lecture notes, with extra explanations and examples, and even a few mathematical derivations—which are not examinable.
3. I will refer to them in class.
4. However, students can ignore them if they don’t find them useful.

**Slide 4-9:** Recall Ch. 2, **Slide 2-19**:

## How the Main Factors Affect Exchange Rates



**Slide 4-15 (Relative PPP)**

From Absolute PPP (slide 4-11), we saw that the following should hold:

$$P_h = e_0 P_f \text{ so that } e_0 = \frac{P_h}{P_f}.$$

If this holds at date 0 and continues to hold, then

at date $t$, we would expect

$$e_t = \frac{P_h(1+i_h)^t}{P_f(1+i_f)^t} = \frac{P_h}{P_f} \times \frac{(1+i_h)^t}{(1+i_f)^t} = e_0 \times \frac{(1+i_h)^t}{(1+i_f)^t}.$$

**Aside: a math result:**

Suppose $\dfrac{1+x}{1+y}=1+a$ and $a\times y\approx 0$. Then, cross-multiplying gives

$$1+x=(1+a)(1+y)=1+a+y+a\times y\approx 1+a+y.$$

So, $a\approx x-y$. But $a=\dfrac{1+x}{1+y}-1$ by assumption, which gives

$$\frac{1+x}{1+y}-1\approx x-y.$$

**Slide 4-18:** For $t = 1$,

$$\frac{e_t-e_0}{e_0}=\frac{1}{e_0}\left(e_t-e_0\right)=\frac{1}{e_0}\left(e_0\times\frac{(1+i_h)}{(1+i_f)}-e_0\right)=\frac{(1+i_h)}{(1+i_f)}-1\approx i_h-i_f.$$

**Slide 4-24: Real Rate of Interest, Nominal Rate of Interest, and Inflation**

**Example:** A can of Coke costs $p_0$ = \$2.00 at time 0, and with inflation it is expected to cost $\tilde{p}_1$ = \$2.10 in one year, i.e., *the inflation rate* is

$$i = \frac{\tilde{p}_1 - p_0}{p_0} = 5\%.$$

The current 1-year *nominal interest rate* is $r$ = 9%. If you buy 100 Cokes today, you will pay \$200. If you invest the \$200 for one year, you will have \$218 and since Coke will cost \$2.10 per can, you can buy 218/2.10 = 103.81 cans. So, we say that the *real* interest rate is $a$ = 3.81%; the formula is $1 + a = \dfrac{1+r}{1+i}$. Also, on slide 4-26, we write this as $(1+r) = (1+a)(1+i)$.

This is the Fisher Effect.

From our maths result, $a = \frac{1+r}{1+i} - 1 \approx r - i$.

## Slide 27

Suppose real returns are equal across countries: $a_h = a_f$.

This means that

$$\frac{1+r_h}{1+i_h} = 1+a_h = 1+a_f = \frac{1+r_f}{1+i_f}.$$

Or, rearranging:

$$\frac{1+r_h}{1+r_f} = \frac{1+i_h}{1+i_f}.$$

NOTE: on slide 4-15, Relative PPP implied: $$e_1 = e_0 \times \frac{1+i_h}{1+i_f}.$$

By the above equality, we have, $$e_1 = e_0 \times \frac{1+r_h}{1+r_f}.$$

**Slide 4-29**

From the equation $\frac{1+r_h}{1+r_f} = \frac{1+i_h}{1+i_f}$ (on slide 4-27, which assumes real returns are equal across countries) and our maths result,

$$r_h - r_f \approx \frac{1+r_h}{1+r_f} - 1 = \frac{1+i_h}{1+i_f} - 1 \approx i_h - i_f .$$

**EXTRA COMMENTS on Slide 4-29:** From Slide 4-27,

$\frac{1+r_h}{1+i_h} = 1 + a_h \approx 1 + a_f = \frac{1+r_f}{1+i_f}$. Subtract 1 from each term:

$$r_h - i_h \approx \frac{1+r_h}{1+i_h} - 1 = a_h \approx a_f = \frac{1+r_f}{1+i_f} - 1 \approx r_f - i_f .$$

At point D: $r_h - r_f = -3; i_h - i_f = -2$, or

$$\left.\begin{aligned} r_h + 3 &= r_f \\ i_h + 2 &= i_f \end{aligned}\right\}. \text{ Subtracting} : r_h - i_h + 1 = r_f - i_f .$$

So, the foreign country has a *higher* real interest rate and so funds should flow from the home country to the foreign country.

**Slide 4-35**

Let's show that $\overline{e}_t = e_0 \times \frac{(1+r_h)^t}{(1+r_f)^t}$.

From Relative PPP (slide 4-15), $e_t = e_0 \times \frac{(1+i_h)^t}{(1+i_f)^t}$.

Assuming real interest rates are equal across countries (slide 4-27):

$$\frac{(1+r_h)}{(1+r_f)} = \frac{(1+i_h)}{(1+i_f)}.$$

This implies that $e_t = e_0 \times \frac{(1+r_h)^t}{(1+r_f)^t}$, but we write it with $\overline{e}_t$:

$\overline{e}_t = e_0 \times \frac{(1+r_h)^t}{(1+r_f)^t}$, which is IFE. (This will relate to *forward prices*, too.)

**Slide 4-36** (word processing glitch)

Example: $r_{UK} = 8\%$, $r_{US} = 6\%$, $e_0 = \text{USD}\,1.6545\,/\,\text{GBP}$.

$$\overline{e}_t = e_0 \times \frac{(1+r_h)^t}{(1+r_f)^t} = e_0 \times \frac{(1+r_{US})^t}{(1+r_{UK})^t} = 1.6545 \times \frac{1.06}{1.08} = 1.6239 < e_0.$$

## Slide 4-37

Here, we want to show that $\frac{\overline{e}_t - e_0}{e_0} \approx r_h - r_f$. This is very similar to what we saw for PPP: $\frac{e_t - e_0}{e_0} \approx i_h - i_f$. For $t = 1$,

$$\frac{\overline{e}_t - e_0}{e_0} = \frac{1}{e_0}\left(\overline{e}_t - e_0\right) = \frac{1}{e_0}\left( e_0 \times \frac{(1+r_h)}{(1+r_f)} - e_0 \right) = \frac{(1+r_h)}{(1+r_f)} - 1 \approx r_h - r_f,$$

where the last step is our maths result, again.

**EXTRA COMMENT: Slide 4-38**

At point F,

$$\frac{\overline{e}_t - e_0}{e_0} = 3\% \text{ and } r_h - r_f = 2\%.$$

If we invest in the home country we get a 2% higher return.

If we buy the foreign currency and invest, we get a lower return, but we get a 3% return on the currency.

So, invest in the foreign currency.

**Slide 4-39**

A **forward exchange contract** is a private agreement to buy or sell a specified amount of foreign currency at a fixed rate on a future date, allowing firms to lock in exchange rates and eliminate FX risk. These contracts trade **over-the-counter**, mainly through banks, and are widely used by corporates and financial institutions for hedging rather than speculation.

In general, futures are traded on exchanges, but forwards are over-the-counter.

One of the key differences between forwards and futures is that futures are “marked to market”.

**Slide 4-44**

We're investing at (date $t = 0$) USD 1 million for 3 months (date $t$).

Spot rate is $e_0 = \text{EUR } 0.74000 / \text{USD}$. Here, EUR is the home currency!

3-month forward rate is

$$f_3 = \text{EUR } 0.73637 / \text{USD} = \text{price of USD in terms of EUR.}$$

US interest rate is 8% per year, or $r_{US} = 2\%$ per 3-month period.

European interest rate is 6% per year or $r_{EUR} = 1.5\%$ per 3-month period.

**Alternative investment**: Invest in US: future value is

$$\text{USD 1 million} \times \left(1 + r_{US}\right) = \text{USD 1,020,000.}$$

1. Convert USD 1,000,000 to EUR to Invest in Europe:

   Euros = $1{,}000{,}000 \times e_0$ = EUR 740,000 (at date 0).

2. Invest at 6% or $r_{Eur} = 1.5\%$ per 3-month period

   Future value is EUR $740{,}000 \times \left(1 + r_{EUR}\right)$ = EUR $751{,}100$.

3. Convert to USD at forward rate: $\dfrac{751{,}100}{f_3} = \dfrac{751{,}100}{0.73637} = \text{USD}\, 1{,}020{,}004$.

Summary of cashflow calculations:

$$\frac{1{,}000{,}000 \times e_0 \times \left(1 + r_{EUR}\right)}{f_3} = \text{USD } 1{,}020{,}004$$

$$1{,}000{,}000\left(1 + r_{US}\right) = 1{,}020{,}000 \approx 1{,}020{,}004.$$

In general (ignoring the 1,000,000), we must have (ignoring the size of the initial cash-flow):

$$\frac{e_0 \times \left(1 + r_{EUR}\right)}{f_3} = \left(1 + r_{US}\right), \text{ or}$$

$$f_t = e_0 \times \frac{\left(1 + r_{EUR}\right)^t}{\left(1 + r_{US}\right)^t} = \text{forward price of USD in terms of EUR.}$$

In general: $$f_t = e_0 \times \frac{\left(1 + r_h\right)^t}{\left(1 + r_f\right)^t}.$$ This is **Interest Rate Parity** (IRP).

Compare to IFE: $$\overline{e}_t = e_0 \times \frac{\left(1 + r_h\right)^t}{\left(1 + r_f\right)^t}.$$

**Slide 4-50**

Let's re-write the no-arbitrage condition as: $f_t \times \left(1+r_f\right)^t = e_0 \times \left(1+r_h\right)^t$.

Now let's assume this *doesn't* hold. Suppose

$$f_t \times \left(1+r_f\right)^t > e_0 \times \left(1+r_h\right)^t$$

(high side) (low side)

**Intuition:** to take advantage of this arbitrage opportunity, date $t = 0$:

$e_0$ is on the "low side", so buy the foreign currency;

$f_t$ is on the "high side" so sell (short) the futures;

$r_h$ is on the "low side", so *borrow* in the domestic market;

$r_f$ is on the "high side", so *invest* in the foreign market.

(Can you do the $f_t \times \left(1+r_f\right)^t < e_0 \times \left(1+r_h\right)^t$ case on your own?)

**Slide 4-50 (cont'd)**

In this case, $r_h$ = 7%, $e_0$ = USD 1.95/ GBP, $r_f$ = 12%, $f_1$ = USD 1.87/ GBP, and $t$ = 1.

$$f_1 = 1.87 > 1.95 \times \frac{(1+0.07)}{(1+0.12)} = 1.8629, \text{ so } f_t \times (1+r_f)^t > e_0 \times (1+r_h)^t.$$

Here are the transactions. Note that #forwards = (1.12)×1,000,000/1.95

| Date $t$ = 0 | Date $t$ = 1 |
|---|---|
| Borrow \$1,000,000 at 7% | Pay \$1,070,000 |
| Buy 1,000,000/1.95 GBP | |
| Invest GBP 512,820.51 at 12% | Receive GBP 512,820.51×(1.12) |
| | Sell GBP 574,385.97 at USD 1.87/ GBP |
| | = Receive \$1,074,051.28 |
| Sell GBP 574,385.97 forwards | |

Note: (a) 1,000,000/1.95 = 512,820.51; (b) 512,820.51×(1.12) = 574,385.97; and (c) 574,385.97×1.87 = 1,074,051.28.

**Slide 4-50 (cont'd) [READ ON YOUR OWN]**

Let's repeat the previous example, except assume $f_1$ = USD 1.85/ GBP.

In this case,

$$f_1 = 1.85 < 1.95 \times \frac{(1+0.07)}{(1+0.12)} = 1.8629, \text{ so } f_t \times (1+r_f)^t < e_0 \times (1+r_h)^t.$$

Here are the transactions:

| Date $t$ = 0 | Date $t$ = 1 |
|---|---|
| Borrow GBP 512,820.51 at 12% | Pay GBP 512,820.51×(1.12) |
| Sell 512,820.51 GBP for $1M. | |
| Invest $1,000,000 at 7% | Receive $1,070,000 |
| | Buy GBP 574,385.97 at USD 1.85/ GBP |
| | = Pay $1,062,614.04 |
| Buy GBP 574,385.97 forwards | |

Note: (a) 1,000,000/1.95 = 512,820.51; (b) 512,820.51×(1.12) = 574,385.97; and (c) 574,385.97×1.85 = 1,062,614.04.

**Slide 4-60**

The equation for $e_5$ is just the equation for the forward rate that we derived based on Slide 4-44.

First, let's re-write the numbers using general notation:

5-year USD interest rate is 6% = $r_{US}$; and 5-years = $t$.

5-year EUR interest rate is 5% = $r_{EUR}$.

Current exchange rate is USD 0.90/EUR = $e_0$.

Value of EUR in 5 years = $e_t = f_t$.

Here is the equation on Slide 4-60:

$\frac{(1.05)^5 \times e_5}{0.90} = (1.06)^5$, which becomes $\frac{(1 + r_{EUR})^t \times f_t}{e_0} = (1 + r_{US})^t$.

Solving for the forward rate gives Interest Rate Parity (IRP):

$$f_t = e_0 \times \frac{(1 + r_{US})^t}{(1 + r_{EUR})^t} = \text{forward price of EUR in terms of USD.}$$

## EXTRA COMMENT: Slides 4-52 and 4-53

The vertical axis says

Interest differential in favour of home country, i.e., $r_h - r_f \approx \dfrac{1+r_h}{1+r_f} - 1$;

The horizontal axis says

"*Forward premium (+) or discount (−) on foreign currency (%)*"

The forward premium is $\dfrac{f - e_0}{e_0}$.

At Point G, $r_h - r_f \approx \dfrac{1+r_h}{1+r_f} - 1 = 2\%$, while $\dfrac{f - e_0}{e_0} = 3\%$, so

$$\frac{f - e_0}{e_0} > \frac{1+r_h}{1+r_f} - 1.$$

After some algebra, we see that

$$f - e_0 > e_0 \frac{(1+r_h)}{(1+r_f)} - e_0, \text{ or } f > e_0 \frac{(1+r_h)}{(1+r_f)}.$$

Here, you buy the foreign currency and sell the forward (so that the exchange rate risk is covered.) With the numbers given on slide 4-52, we will make an extra 1% annually.

**Slides 4-52 and 4-53 (cont'd)**

At Point H, $r_h - r_f \approx \frac{1+r_h}{1+r_f} - 1 = 4\%$, while $\frac{f - e_0}{e_0} = 3\%$, so

$$f < e_0 \frac{\left(1+r_h\right)}{\left(1+r_f\right)}.$$

Here, you sell the foreign currency and buy the forward (so that the exchange rate risk is covered.) With the numbers given on slide 4-52, we will again make an extra 1% annually.